

Supplementary Historical SIR-ODE Benchmark

These historical results are retained for reference only; they do not support the power-grid conclusions. SIR-ODE uses a population of 500, two initial infections, 100 candidate measurement times, observation noise SD 1.0, and a precomputed simulation bank. The objective remains EIG, estimated here by averaging prior-to-posterior entropy reductions over a 1,024-particle representation. This estimator has ceiling $\log(1024) = 6.9315$ nats; the grid experiments instead use sPCE.

Table 1 reports EIG as mean $\pm$ sample SD in nats. DAD/RL-sBOED/Step-DAD use three training-seed means; Random/Myopic/Fixed use five evaluation-seed means after deduplication. Each evaluation uses 512 systems; Random averages 32 design replicates per system. Fixed is calibrated once. Historical training uses a behavioral-cloning warm start, different optimizers, and checkpoint selection with a sequence-diversity preference. Full settings, models and results remain in the companion collection.

Table 1: SIR-ODE EIG

Method	$T = 3$	$T = 4$	$T = 5$
Random	3.7783 ± 0.0167	3.8593 ± 0.0167	3.9241 ± 0.0167
Fixed	6.2816 ± 0.0101	6.4501 ± 0.0091	6.5360 ± 0.0044
Myopic	6.7108 ± 0.0125	6.7437 ± 0.0124	6.7596 ± 0.0135
DAD	6.8394 ± 0.0087	6.8591 ± 0.0039	6.8658 ± 0.0077
RL-sBOED	6.8494 ± 0.0029	6.8674 ± 0.0020	6.8814 ± 0.0015
Step-DAD	6.8438 ± 0.0095	6.8679 ± 0.0011	6.8761 ± 0.0109